JAYCE HILL

Principal Security Engineer, AI | Agentic AI Security · Runtime & Identity · Threat Modeling at Scale

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EXECUTIVE SUMMARY

Principal security architect with 20+ years serving as the accountable security authority for complex distributed systems across Fortune 100 enterprises, the Department of Defense, and the Intelligence Community. Owned the security review and access-control design for **HVE-Core**, a 34-agent AI engineering platform, and co-authored **CRISP**, an MCP-based threat modeling framework deployed across 100+ enterprise environments. Built and open-sourced a **secure MCP execution runtime**: a policy-enforcing proxy with signature-verified, short-lived agent identity, plus a gVisor execution-sandbox design. Brings formal accreditation experience translating regulatory requirements into engineering controls, plus hands-on offensive security and incident response.

CORE COMPETENCIES

- AI Platform & Inference Security
- Runtime & Multi-Tenant Isolation (gVisor sandbox design)
- Prompt Injection & Adversarial Input Defense
- Model & Software Supply Chain (SBOM, SLSA, Sigstore)
- Threat Modeling at Scale (CRISP, STRIDE)
- AuthN/AuthZ & Non-Human Identity
- Security Architecture Reviews & Standards
- Compliance-to-Controls Translation (RMF, NIST 800-53)
- Incident Response & Post-Incident Analysis
- Offensive Security (MITRE ATT&CK, purple team)
- Cloud & Container Security (Azure, AKS)
- Policy-as-Code & Automated Enforcement

TECHNICAL SKILLS

AI Security: OWASP Top 10 for LLM, NIST AI RMF, Google SAIF; MCP security proxy, OPA, Entra Agent ID, gVisor, Defender for AI
Cloud & Platform: Microsoft Azure (Solutions Architect Expert), AKS, AI Foundry, Key Vault, Defender for Cloud, Sentinel; situational AWS and OCI
Supply Chain & CI/CD: CodeQL, GitHub Advanced Security, Trivy, CycloneDX, Sigstore/Cosign, SLSA, OpenSSF Scorecard
Languages / IaC: Python, PowerShell, Bash, Terraform, Bicep

PROFESSIONAL EXPERIENCE

Principal Security Software Engineer | Microsoft, Industry Solutions Engineering *2021 – 2026*

Forward Deployed Engineer embedded with Fortune 100 and public sector customers; Security Lead for ISE's agentic AI engineering platform. Individual contributor, Level 65 (Principal).

- Owned the security review for **HVE-Core**, a production-scale 34-agent AI engineering platform: co-designed the access-control architecture, threat-modeled agent permissions and trust boundaries, and red-teamed agents to enforce human-in-the-loop and accuracy controls before release.
- Co-authored and co-led **CRISP**, a threat modeling framework operating as an MCP server that scans architectures and generates security plans; deployed across 100+ enterprise environments globally and named a top-priority accelerator by ISE leadership.
- Conducted architecture and code reviews across 20 personally led engagements (including Bank of America, Fidelity, LSEG, General Motors, and KPMG), driving recurring findings to root cause and standardizing fixes into CRISP's frameworks.
- Authored a secure-by-default Bicep library spanning 11 Azure services (AKS, SQL, Cosmos DB, Key Vault, Redis, AI Foundry, Storage, VNet, NSG, PIM) with Zero Trust defaults, and hardened CRISP's CI/CD and software supply chain (CodeQL SAST, GitHub Actions SHA pinning, Trivy, CycloneDX SBOM signing) to OpenSSF Scorecard standards.
- Translated Defender for Cloud findings into policy-as-code deployed directly into customer environments, automating enforcement of security baselines.

Information System Security Manager | General Dynamics IT (GDIT) *2020 – 2021*

Security accreditation and operations for classified cloud systems supporting IC missions.

- Owned security accreditation (ATO) and Continuous Monitoring under RMF for classified AWS-based systems, translating control requirements into engineering implementations with the Authorizing Official.
- Led post-incident analysis for data spills and drove zero-day vulnerability resolution; ran architecture reviews to set controls and remediation milestones.

Senior Principal Consultant, National Security Group | Oracle *2011 – 2020*

Built and implemented offensive and defensive cyber capabilities for DoD customers over a 9-year engagement.

- Led MITRE ATT&CK-based penetration testing, purple team operations, and vulnerability assessment across Oracle Exadata, Big Data Appliance, WebLogic, and Analytics platforms, validating adversary tradecraft against DoD mission systems.
- Served as Oracle Cloud Infrastructure Solution Architect, designing deployment architectures and leading certification and accreditation for national security programs; translated technical findings into program- and executive-level risk posture.
- Facilitated quarterly stakeholder reviews and bi-monthly Technical Exchange Meetings on risk, emerging technology, and program milestones.

Systems Security Engineer | Booz Allen Hamilton *2008 – 2011*

- Performed certification and accreditation, risk assessments, and security control validation for NSA mission systems under DCID 6/3, coordinating findings with government stakeholders and program security officers.

Earlier Career | NJVC · L-3 · Lockheed Martin · Titan Systems *2003 – 2008*

- Systems and security engineering roles supporting national security and defense programs.

INDEPENDENT RESEARCH & HANDS-ON LABS

Secure MCP Execution Runtime — Open-Source Reference Architecture | Hill Security Advisors *2026*

- Built and open-sourced a secure execution runtime for agentic AI tool calls: a Python JSON-RPC security proxy with recursive argument sanitization and structured telemetry, a proof-of-possession (RFC 7800) token validator for short-lived agent identities, and a default-deny OPA/Rego tool-authorization policy, with unit tests and CodeQL running in CI.
- Delivered Azure IaC for the control plane (Entra Agent ID, APIM inspection policy, KQL audit queries), a threat matrix mapping prompt injection, excessive agency, and non-human identity risk to OWASP Top 10 for LLM, Google SAIF, and NIST AI RMF, and a gVisor execution-sandbox design; code at github.com/jaycehilleng/jaycehilleng.github.io.

OpenClaw Agentic AI Lab | Hill Security Advisors *2026*

- Building and security-reviewing OpenClaw, an open-source agentic AI assistant, on an isolated Azure VM in a dedicated lab tenant; designing exec-approval routing and human-in-the-loop gating before enabling autonomous tool execution.

EDUCATION

M.S. Cybersecurity (in progress) — Georgia Institute of Technology
M.Eng., Systems Engineering — George Washington University
B.S., Electrical & Computer Engineering — Michigan State University

CERTIFICATIONS & CLEARANCE

Microsoft Azure Solutions Architect Expert · GIAC Security Essentials (GSEC) · Certified Ethical Hacker (CEH)
TS/SCI — Active (Full Scope Polygraph held continuously 2003–2023)

COMMUNITY LEADERSHIP

100 Black Men of Greater Washington, DC: Vice President of Operations and Development (2015–2019); Board of Directors (2011–2019). Created STEM The 100 Way, impacting 3,000+ students, including an ongoing STEM partnership with the Department of Defense.